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Machine learning prediction of physical properties and nitrogen content of porous carbon from agricultural wastes: Effects of activation and doping process

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ABSTRACT

In the study, machine learning models for prediction specific surface area (SSA), total pore volume (TPV), microporosity (MP) and nitrogen content (NC) were developed based on three tree-based models by collecting published experimental data on the agricultural wastes for the preparation of nitrogen-doped porous carbon, in which the physicochemical properties of raw materials and activation conditions were considered and compared as input features. By randomly combining different groups of input features in the three models, gradient boosted decision tree (GBDT) with the combination of all features presented the highest test R^2 of 0.939, 0.835, 0.861 and 0.906 for the four prediction targets with high accuracy and good generalization ability. The visual analysis of the model showed that the activation temperature and impregnation ratio were the most critical factors for the targets, while the NC was additionally influenced by the nitrogen-doping ratio. The SSA, TPV and MP peaked around the activation temperature of 800 °C and the impregnation ratio of 3, while NC also increased with the amount of nitrogen doping and reached saturation of nitrogen fixation at the nitrogen-doped ratio of 3. This data-driven scientific tool effectively guided the synthesis of porous carbon and broadened the utilization of biomass energy.

1. Introduction

Biomass has drawn much attention as one of the most plentiful and promising renewable feedstocks in response to the simultaneous challenges of rising global energy demand and environmental pollution by fossil fuels [1]. An estimated 3.6 billion tons of agricultural wastes is produced annually, in which a substantial number is lignocellulosic biomass [2]. The two primary methods for converting lignocellulosic biomass into energy, fuels and bioproducts are biochemical and thermochemical technology, in which the thermochemical processes like pyrolysis, gasification and hydrothermal liquefaction proceed fast and efficient, lots of the researchers' attention have been focused on it in recent years [3]. Pyrolysis is a cutting-edge method for converting biomass into bio-oil, gaseous products and biochar in the oxygen-restricted environment [4].

Biochar, an essential byproduct of biomass pyrolysis, should be further activated, modified or subjected to other treatments to create a variety of high-performance carbon materials that can be used in catalysis, adsorption and energy storage, which plays a significant role

in widening the efficiency and adaptability of biomass pyrolysis [5]. Two different ways are usually utilized in the activation of biochar, namely, physical and chemical activation. In the physically activation, Steam, nitrogen and carbon dioxide are primarily used to reduce the carbonaceous materials gently. And in the chemical activation process, a variety of activating agents, including KOH, ZnCl₂, NaOH and H₃PO₄ are used [6]. By combining the carbonization and activation processes in one step, chemical activation presents great advantages over physical activation in terms of energy savings, since it occurs easily at a lower temperature. Other key advantages of the chemical activation technique include higher carbon yields, shorter activation times, appropriate surface functional groups and higher activated carbon surface area [7]. In previous studies, the application of porous carbon can be significantly impacted by heteroatom doping. The crystal and electronic structure of the porous carbon may be altered by the addition of heteroatoms (N, O, P and B), impacting the electrolyte's capacitive capabilities significantly and resulting in capacitive reactions [8]. The surface electron distribution of porous carbon was altered by nitrogen loading and the nitrogen-containing functional groups (pyridinic-N, pyrrolic-N, graphitic-N and

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oxidic-N) are created, causing base-acid interactions between nitrogen functional groups and CO₂ molecules, and finally the stability of CO₂ adsorption of porous carbon is promoted [9].

The physicochemical properties of nitrogen-doped porous carbon depend on various factors. The characteristics of various carbon precursors, especially some specific biomass with inherent structural advantages presented significant influence on the physicochemical characteristics of porous carbon. Water hyacinth stems with the thin layer structure was selected as the carbon source by Wu et al., and hemicellulose was firstly removed by acid washing, and it was carbonized followed by KOH activation, obtaining the porous carbon with a desirable pore volume of 0.84 cm³/g and a specific surface area of 1308 m²/g [10]. The physicochemical property of porous carbon may also be significantly impacted by activation temperature and time. Liu et al. prepared nitrogen-doped porous carbon using CaCl₂ as the activator and melamine as the nitrogen source, and the prepared samples showed very different layered porous structures and nitrogen contents in the activation temperature range of 600 ~ 800 °C. The samples activated at 800 °C exhibited high specific surface area (892 m²/g), appropriate nitrogen doping (5.63 wt%), numbers of micropores and uniform mesopores. [11]. Also, results can vary depending on the type of activating agents and the amount of activating agents. Among several popular activation agents, KOH often frequently used, by which the enlargement of micropores easily occur, and smaller mesopores were produced with ZnCl₂, and the uneven pore size distribution could be obtained with H₃PO₄ as activation agent. By using willow wool as the raw material to create nitrogen and sulfur co-doped porous carbon, Li et al. found that porous carbon with an excellent specific surface area of 1431 (m²/g) was produced at an impregnation ratio of 2. In addition to the above listed variables, other factors including the nitrogen-doped ratio, heating rate and atmosphere played the significant role in the preparation of porous carbon. To obtain the optimal process parameters for producing nitrogen-doped porous carbon from biomass, numerous and repeated tests are often needed. To reduce the economic and time costs, the modeling of biomass thermal conversion process is crucial.

As artificial intelligence and big data evolve quickly, data-driven machine learning (ML) has been extensively utilized in the field of biomass thermal conversion [13], like biochar synthesis and its adsorption on heavy metal ions and organic compounds. Based on the biomass pyrolysis data, The yield of biochar and carbon content of porous carbon was successfully predicted by Zhu et al. with biomass characteristics and pyrolysis conditions as input features using RF model, and the effect of pyrolysis conditions on the objectives was also explored. [15]. The industrial composition and elemental composition of organic waste pyrolysis biochar predicted by Li et al. using an Artificial Neural Network (ANN) and the great dependence of these features was explained [16]. Several popular machine learning (ML) models are often used, including ANN, Support Vector Machines (SVM) and linear regression [17]. The tree-based machine learning models is a subset of supervised machine learning techniques, in which binary segmentation of data is often used to minimize the residual sum of squares [13]. Light Gradient Boosters (LGB), Decision Trees (DT), Random Forests (RF), Gradient Boosted Decision Trees (GBDT) and Extreme Gradient Boosting (XGB) were most popular and commonly used tree-based machine learning models, in which the last three models have attracted a lot of interest in scientific research due to their enhanced over-fitting resistance, superior noise resistance and applicability in handling smaller data sets [18]. GBDT and XGB are more substantially advantages over commonly used RF by distributing unbiased weights to relevant features and efficiently handling unbalanced datasets [19,20]. In the substantial prior research, the boosted-tree technique enabled faster hyperparameter tuning than the more popular SVM and ANN algorithms when using small data sets (300–1000 data points) without sacrificing accuracy.

ML prediction of nitrogen-doped porous carbon prepared by biomass carbonation followed by activation have been little done by researchers.

In this work, the prediction of the specific surface area (SSA), total pore volume (TPV), microporosity (MP) and nitrogen content (NC) of porous carbon was done, while taking the industrial composition, elemental composition, biochemical composition, extra chemicals addition and activation condition as the input feature considering that both the biomass fast pyrolysis and the activation process are all multivariate nonlinear processes, the prediction performance of multiple learning algorithms including GBDT, RF and XGB, was evaluated. The model was optimized in two ways, input features selection and hyperparameters adjustment. In addition, the effects of biomass properties, chemicals and activation process in the preparation of porous carbon were investigated by characteristic contribution and partial dependence plot (PDP) analysis. Using machine learning models contributed to a more profound knowledge of the activation and doping process, providing guiding proposals for controlling the physicochemical characteristics of the nitrogen-doped porous carbon.

2. Methods

2.1. Data collection and preprocessing

The database in this study was collected from published literature about the conversion of lignocellulosic agricultural wastes (including various wood wastes, corn husks, corn cobs, wheat straw, rice straw and sugarcane bagasse) into nitrogen-doped porous carbon by two-step pyrolysis and activation method. The literature was obtained by searching on various online website such as Science Elsevier, Web of Science, Scopus and Google Scholar with the keywords like N-doped, carbon, agricultural waste, BET and activation. In this study, the experimental methods were restricted to a two-step preparation approach of pyrolysis and activation, with great emphasis on the activation process. To ensure that the physicochemical property of the pyrolysis biochar produced in the first step of each literature was approximately the same, the process parameters of the pyrolysis process was concentrated within in the pyrolysis temperature of 400–500 °C, the pyrolysis time of 1–2 h and the heating rate of 5–10 °C/min in the collected literature. Several activating agents commonly used by researchers (including KOH, ZnCl₂, H₃PO₄ and NaOH, etc.) and two additional sources of nitrogen doping (including melamine and urea) was restricted in this study. For the literature with some missing raw material characteristics, we used the biomass open-source website Phyllis2 (<https://phyllis.nl/>) to supply relevant data. In addition, the mean value of the entire column of features is used to supplement the still missing characteristics of the raw material. Finally, 49 published papers satisfied our requirements for establishing the required database. Plot Digitizer 2.6.8 software (<http://plotdigitizer.sourceforge.net/>) was used to identify and collect data from plots or read directly from tables. The final data collected included SSA (253 sets of data), TPV (258 sets of data), MP (243 sets of data) and NC (162 sets of data). Details of the datasets are provided in [Supplementary Material \(Table S1\)](#).

2.2. Selection of input features

In total, the input characteristics can be divided into five groups. (I) Elemental composition of biomass, including the relative content of carbon (C), hydrogen (H), oxygen (O) and nitrogen (N), each of which had significant effects on the pyrolysis products. (II) Industrial composition of biomass, including moisture (M), ash (Ash), volatiles (V) and fixed carbon (FC), in which moisture was measured before the feedstock dried. (III) Biochemical composition of biomass, including the relative content of cellulose (Cel), hemicellulose (Hem) and lignin (Lig), ignoring the other constituents. (IV) Chemicals, including the ratio of activating agent to biocarbon and the ratio of nitrogen-doped source to biocarbon, noted as impregnation ratio (IR) and nitrogen-doped ratio (DR). (V) Activation conditions, including activation temperature (AT), heating rate (HT) and residence time (RT), where the activation temperature had

an important influence on the physicochemical properties of porous carbon and was one of the most critical parameters in the activation process. At lower temperatures, it was not sufficient to form a dense pore structure, while at relatively high temperatures, the surface structure of biochar tended to collapse and the specific surface area and total pore volume decrease. Then, preserving each data subset with two groups of (IV) chemicals and (V) activation conditions, three groups of (I) elemental composition, (II) industrial composition and (III) biochemical composition was combined, obtaining a total of seven subsets, as shown in Table 1.

The elemental and industrial compositions in the data set were based on Eqs. (1), (2), (3) and (4), respectively:

$$O = 100 - C - H - N \quad (1)$$

$$\text{Ash} = a^*(100 - \text{Moisture}) \quad (2)$$

$$\text{FixCarbon} = b^*(100 - \text{Moisture}) \quad (3)$$

$$\text{Volatile} = c^*(100 - \text{Moisture}) \quad (4)$$

where C , H , O and N represented the relative content (wt%) of carbon, hydrogen, oxygen and nitrogen in biomass, respectively; a , b and c represented the relative contents of ash, solid carbon and volatiles, respectively (wt%).

The linear correlation between any two features was described as the Pearson correlation coefficient (PCC), as shown in Eq. (5):

$$PCC = \frac{\sum_{i=1}^n (x_i - \bar{x}) \sum_{i=1}^n (y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (5)$$

PCC is used to detect the collinearity between any two input features and the correlation between the input features and the target features [21]. PCC values was usually in the range of -1 to 1 , in which the positive value indicated a positive correlation, and the negative value indicated a negative correlation. The larger the absolute value of the data, the better the linear relationship between the variables.

2.3. Machine learning models

RF, GBDT and XGB are classified as “Ensemble Learning”, which aims at improving the generalization ability and robustness of a single learner by combining the prediction results of multiple learners.

2.3.1. GBDT model

The GBDT algorithm is trained using the gradient boosting method. First, the leaves are obtained from the average of the observed output values, which is also the initial prediction value for obtaining the initial error [22]. Then, the gradient boosting strategy constructs the first regression tree that predicts the error based on the change between the actual and original values. Next, a new prediction value was obtained by adding the previous prediction value to the prediction error of the new regression tree [23]. Finally, the next regression tree is constructed based on the new residuals of the predicted and actual experimental

values of the previous tree until the residuals are minimized.

2.3.2. RF model

Random Forest (RF) is an extended variant of Bagging, which further introduces random feature selection in the training process of decision trees based on the decision tree as the base learner to build Bagging integration, so it can be summarized that RF includes four parts: firstly, random sample selection (put-back sampling); secondly, random feature selection; Thirdly, construction of decision trees; Fourthly, random forest voting (averaging). Random selection of features means that in the tree construction, some features are randomly selected from the set of features in the sample set, and then the best attributes are selected from this subset for division. However, the “average” property of the random forest will make its variance decrease, and the decrease in variance compensates for the increase in bias, so it is a better model overall. Each of RF’s decision trees is maximally grown without pruning when constructing decision trees.

2.3.3. XGB model

XGB is an integrated algorithm for minimizing residual errors with fast iteration speed and the ability to handle large scale data. The basis of XGB is the gradient enhancement algorithm, an integrated algorithm model for building predictive models [24]. In this integrated algorithm, multiple weak evaluators are used to model the data, and the results of all weak evaluators are summarized to gain better classification or regression performance [25]. First, a gradient advancement tree is built as the weak estimator, followed by stepwise iterations. The iterations reduce the bias to generalization error as a criterion, adding a tree in each iteration, and gradually forming a substantial estimator integrated with many tree models. The basic unit of each regression tree is the leaf node, which has a prediction score, also called leaf weight, representing all the examples on this leaf node. The prediction score of the regression ensemble model is the sum tree of all prediction scores [26].

2.4. Model training and evaluation

Tree-based algorithms, such as GBDT, XGB and RF algorithms used in our study, aren’t based on distance metric for model prediction, and aren’t very sensitive to the dimension of features. Therefore, the data wasn’t processed through normalization and standardization. After preprocessing the entire dataset, the data were randomly selected by using 85% of the total data points as the training set and the remaining 15% as the testing set for developing the model and doing the final evaluation. In the training process, 5-Fold cross-validated GirdSaerch method is adopted to optimize the hyperparameters and improved the generalization ability of the model while ensuring the prediction accuracy [27]. In the 5k-fold CV, the training data was divided into five subsets, and in each training iteration, one of the subsets was used as the validation set, and the other four subsets constituted the training set. The determination coefficient (R^2) and the root mean square error (RMSE) of the regression model are used as indicators to evaluate the model. The more accurate and precise the model is, the closer the R^2 is to 1, and the lower the RMSE is, as shown in the following equations

$$R^2 = 1 - \frac{\sum_{i=1}^N (Y_i^{\text{exp}} - Y_i^{\text{pred}})^2}{\sum_{i=1}^N (Y_i^{\text{exp}} - Y_{\text{ave}}^{\text{exp}})^2} \quad (7)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i^{\text{exp}} - Y_i^{\text{pred}})^2} \quad (8)$$

where Y_i^{exp} and Y_i^{pred} are the experimental and predicted values, $Y_{\text{ave}}^{\text{exp}}$ is the mean of the experimental values, and N is the number of samples.

A machine learning model is essentially a black box model. Therefore, it is a challenge to determine the importance or contribution of an input feature to the model. Permutation Importance, Feature

Table 1
Input feature information for the data subset.

Case	Input Combination	Input Variables
1	I, II, III, IV, V	C, H, O, N, M, Ash, V, FC, Cel, Hem, Lig, IR, DR, AT, HT, RT
2	I, IV, V	C, H, O, N, IR, DR, AT, HT, RT
3	II, IV, V	M, Ash, V, FC, IR, DR, AT, HT, RT
4	III, IV, V	Cel, Hem, Lig, IR, DR, AT, HT, RT
5	I, II, IV, V	C, H, O, N, M, Ash, V, FC, IR, DR, AT, HT, RT
6	I, III, IV, V	C, H, O, N, Cel, Hem, Lig, IR, DR, AT, HT, RT
7	II, III, IV, V	M, Ash, V, FC, Cel, Hem, Lig, IR, DR, AT, HT, RT

Importance and SHAP are three commonly used methods to calculate feature importance [28]. The essence of the Feature Importance is that the dependence of features on the trained model does not represent the dependent capability of features in the test set and the verification set. However, the Permutation importance method solves this problem well and can be used on any model. Considering 16 input features in our study, we choose the Permutation Importance method, which can be achieved through the Permutation Importance method in Scikit-Learn. Briefly, it includes altering the data arrangement in one column of the data table while leaving the other features unchanged to see how much it impacts the prediction accuracy. The feature that has the most significant impact on prediction accuracy is the most important feature. Although permutation importance offers useful information about the contribution of each feature to the model, it does not explain how these

input features affect the trend of the output variables. For this reason, once several important input features have been selected, it is necessary to evaluate their variation in relation to the output features, so we also used PDP visualization to help explain the association between each input feature and the target feature [29]. All data processing and machine learning model building was performed in the Python programming language (Version 3.9) using the Scikit-Learn library.

3. Results and discussions

3.1. Statistical analysis of data sets

It began with a preliminary descriptive analysis of the minimum, maximum and concentration points, in which a total of 16 input features

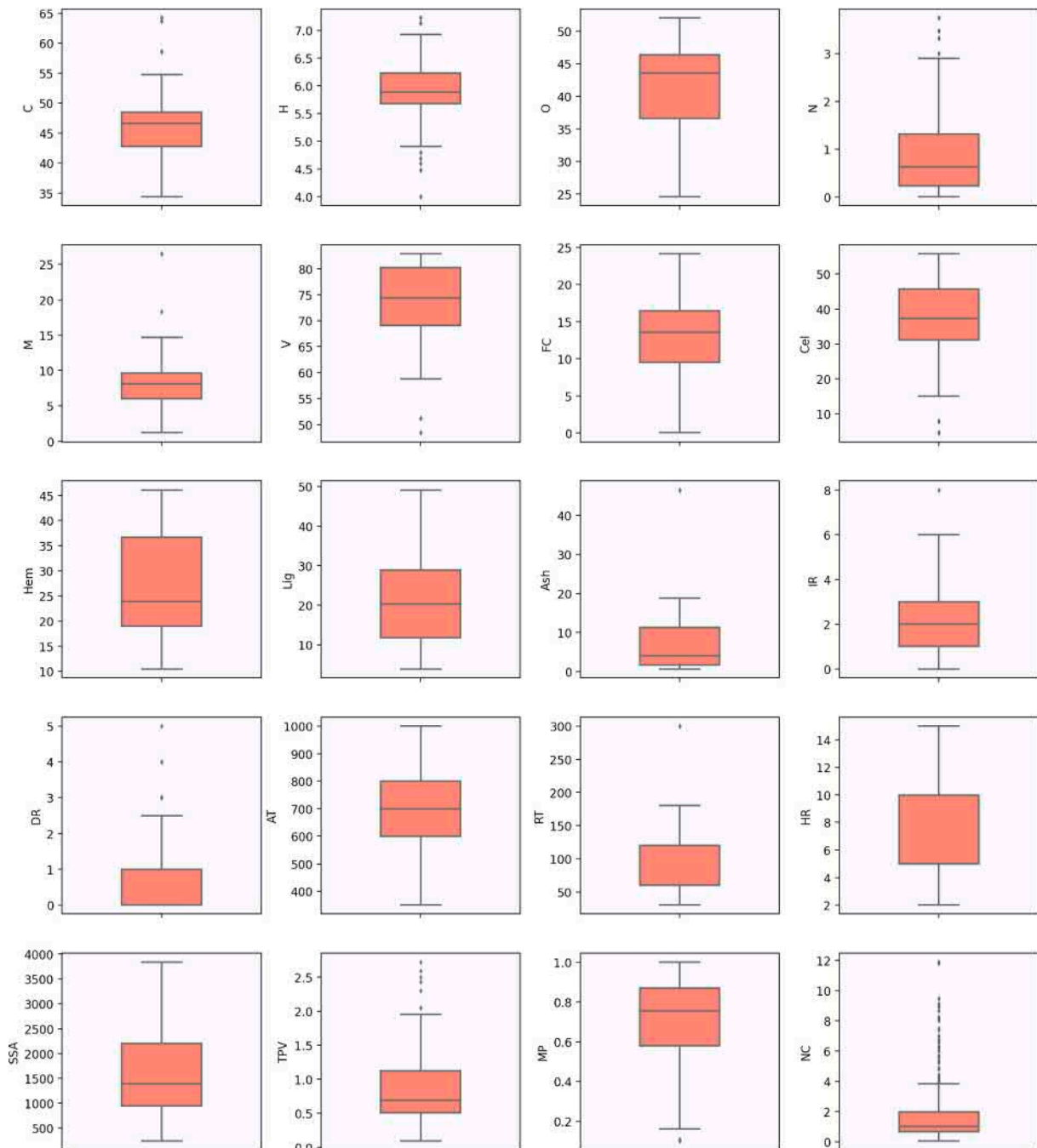


Fig. 1. Box plot of statistical variables related to input and output dataset.

and 4 output features were included. Fig. 1 shows the box plot for each feature. A box plot usually contains six main data nodes, and a set of data is arranged from the largest to the smallest to obtain the upper bound (upper horizontal line), upper quartile Q3 (upper box border), median (middle box line), lower quartile Q1 (lower box border), lower edge (lower horizontal line) and an outlier.

The elemental C, H, O and N in the biomass feedstock range from 34.43%–64.23%, 4%–7.23%, 24.6%–52.04% and 0.1%–3.75%, respectively. The distribution of C and O content is relatively concentrated at 46% and 43%, and the content of H, the basic element of carbohydrates, is concentrated at about 6%. In contrast, the distribution of N content is relatively concentrated at a smaller value, about 0.6%, and the relatively low N content is exactly the characteristic of most lignocellulosic agricultural wastes. The distribution of moisture, ash, fixed carbon and volatiles in the industrial composition of the feedstock range from 1.2%–26.49%, 0.7%–46.49%, 0.04%–24.12% and 48.44%–82.85%, respectively. They were generally concentrated at 8%, 4%, 13% and 74%, which is in general agreement with the data collected in ref [30]. The three feedstock components (cellulose, hemicellulose and lignin) in the lignocellulosic agricultural waste feedstock, in the range of 4.54%–

55.78%, 10.48%–46.08% and 3.92%–49.12% respectively, should also be taken into consideration. These statistics varied relatively in a wide range in each data, which may be related to the diversity and complexity of biomass resources. Among the chemicals, the activating agents play the crucial role for the formation of pores. After limiting several commonly used activating agents, the impregnation ratio as a feature concentrated in the range of 0–6, and it lied on 2 or 3 in most studies. The nitrogen-doped source plays a key role in the preparation of the nitrogen-doped porous carbon. The nitrogen doping ratio distribution was within 0–5. The most crucial activation parameter is the activation temperature, which ranges from 350 °C to 950 °C and is primarily concentrated around 700 °C. The choice of 350 °C is due to the collection of some studies using NaNH_2 as an activating agent, by which the formation of layered pore structures is promoted at lower activation temperature. The heating rate and residence time ranges from 2 to 15 °C/min and 30–300 min. The large variation dimension of various process parameters makes it more challenging for the researchers to determine the optimal working conditions for each test.

In terms of output features, the SSA is distributed in the range of 232–3839 (m^2/g), and most are concentrated around 1400 (m^2/g),

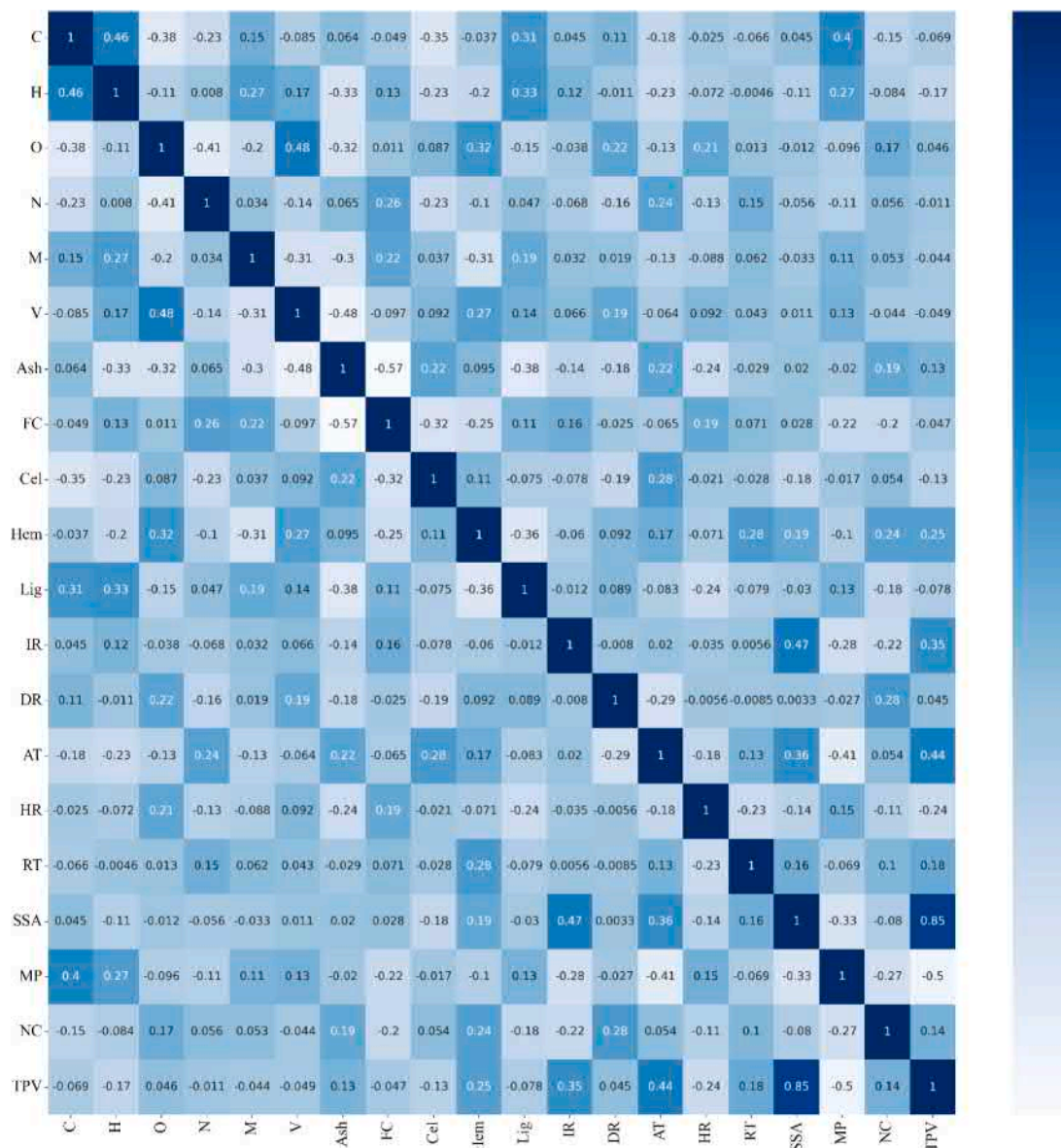


Fig. 2. Pearson correlation matrix among any two variables.

indicating that the prepared porous carbon in most studies are within a large specific surface area. In ref [31], tobacco sticks were first mixed with urea with 1:1 ratio and pre-carbonized at 500 °C for 3 h, and then mixed with KOH with 1:2 ratio and finally activated at 800 °C for 1 h to obtain an ultra-high porous carbon with a SSA of 3839 m²/g. The relative number of micropores and mesopores in the hierarchical porous carbon can be defined by the microporosity to judge whether the target of hierarchical structure is reached. In our collected dataset, the distribution ranges of total pore volume and microporosity are 0.09–2.7 cm³/g and 0.1–0.99.

Fig. 2 shows the PCC relationship between any two variables. First, the TPV is positively associated with the specific surface area. Namely, the high consistency existed between these two, a high SSA of the porous carbon was typically accompanied by a high total pore volume. In terms of the SSA, the impregnation ratio and activation temperature present the slightly larger PCC values of 0.47 and 0.36 than the other features, indicating that these features present a more significant impact on the specific surface area, which has already been a hotspot that attracted the attention of more studies. Similarly, due to the high correlation between TPV and SSA, the impregnation ratio and activation temperature also play a significant role on TPV with high correlation coefficients of 0.35 and 0.44. It further supports the idea that the appropriate design of activation temperature and impregnation ratio was of great significance to successfully achieve pore formation and produce hierarchical porous carbon with well-developed pores. N, O, ASH and FC are highly negatively correlated in their respective groups, and there is no significant correlation among the input features, indicating that the selected features are appropriate to describe the whole modeling process comprehensively and each feature can reflect the corresponding contribution in the model building.

3.2. Comparison of prediction models with various combinations of input features

Fig. 3 provides the Test R² of three models with 7 combination groups of input features. It could be seen that the accuracy of the prediction model fluctuates with various input feature combination groups. Overall, Case 1 performed well in all four target values. As shown in Fig. 3-A aiming at predicting the specific surface area with the GBDT model, the R² decreased from 0.939 in Case1 to 0.913 in Case 2 with only the elemental composition remaining, while the maximum R² appeared in Case4 with only the biochemical composition retained using RF model. In terms of the total pore volume, the prediction accuracy fluctuated widely with 7 combination groups. For example, the R² decreased from 0.862 in Case1 to 0.711 in Case5 using the XGB model, indicating that removing the biochemical composition of biomass feedstock (cellulose, hemicellulose and lignin) helps to filter out some useful information from the modeling process. Zhu et al. [15] also found that the biochemical composition of biomass was more significant than elemental composition in predicting the yield biochar and carbon content. In Fig. 3-C, the Test R² for N content lie in the range of 0.774–0.861 (GBDT), 0.641–0.774 (RF) and 0.764–0.830 (XGB) and were all maximized at Case1, indicating that accuracy of prediction model can be effectively improved by adding relevant input features, and that the more comprehensive input features can be helpful to better describe the prediction model [32,33]. It was worth noting that the GBDT model also achieved good prediction in Case1 (0.861) in terms of microporosity, with a progressively decreasing R² in other combination group and the lowest in Case4 (0.774). The results indicated that Case1 retaining all the features is more comprehensive for building the prediction model, improving the prediction accuracy and the generalization ability to unknown data of the prediction model could be promoted. All discussions followed will be based on Case 1 as the input feature.

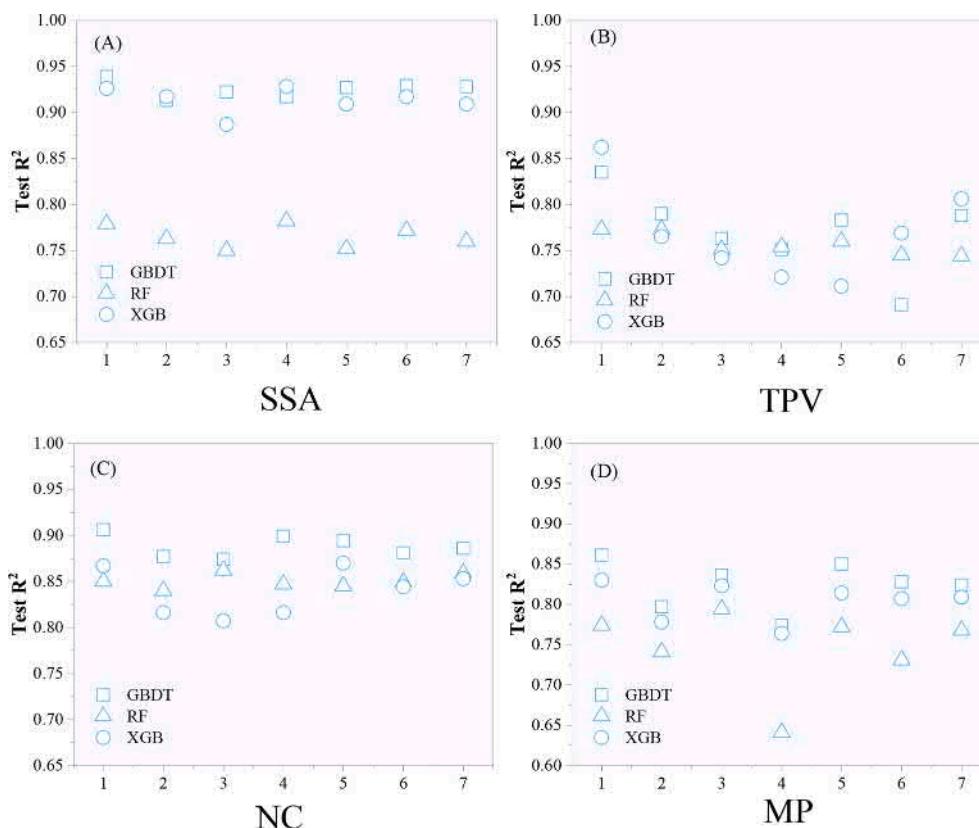


Fig. 3. Test R² of GBDT, RF and XGB with 7 combination groups of input features (a) SSA, (b) TPV, (c) NC and (d) MP.

3.3. Comparison of the prediction accuracy with GBDT, RF and XGB models

The three tree-based ML models, GBDT, RF and XGB were designed and evaluated using the preprocessed data as the input features of the models. The prediction accuracy of models was evaluated by two metrics with training and testing database, R^2 and RMSE. Fig. 4 shows the scatter plot of the actual and predicted values of the four output features (specific surface area, total pore volume, microporosity and N content)

using the best hyperparameters of the three ML models with all input features from Case 1 as the dataset, in which the black dashed line represents $X = Y$, indicating that the predicted value is equal to the actual value, and the closer the point to the $X = Y$ line indicates better prediction performance. The blue shading stands for the 95% confidence interval of the regression line at the test point.

In Fig. 4(a), (b) and (c), it can be seen that almost all points of the training and testing sets in terms of the specific surface area are concentrated on the $X = Y$ line. However, the density of the points in the

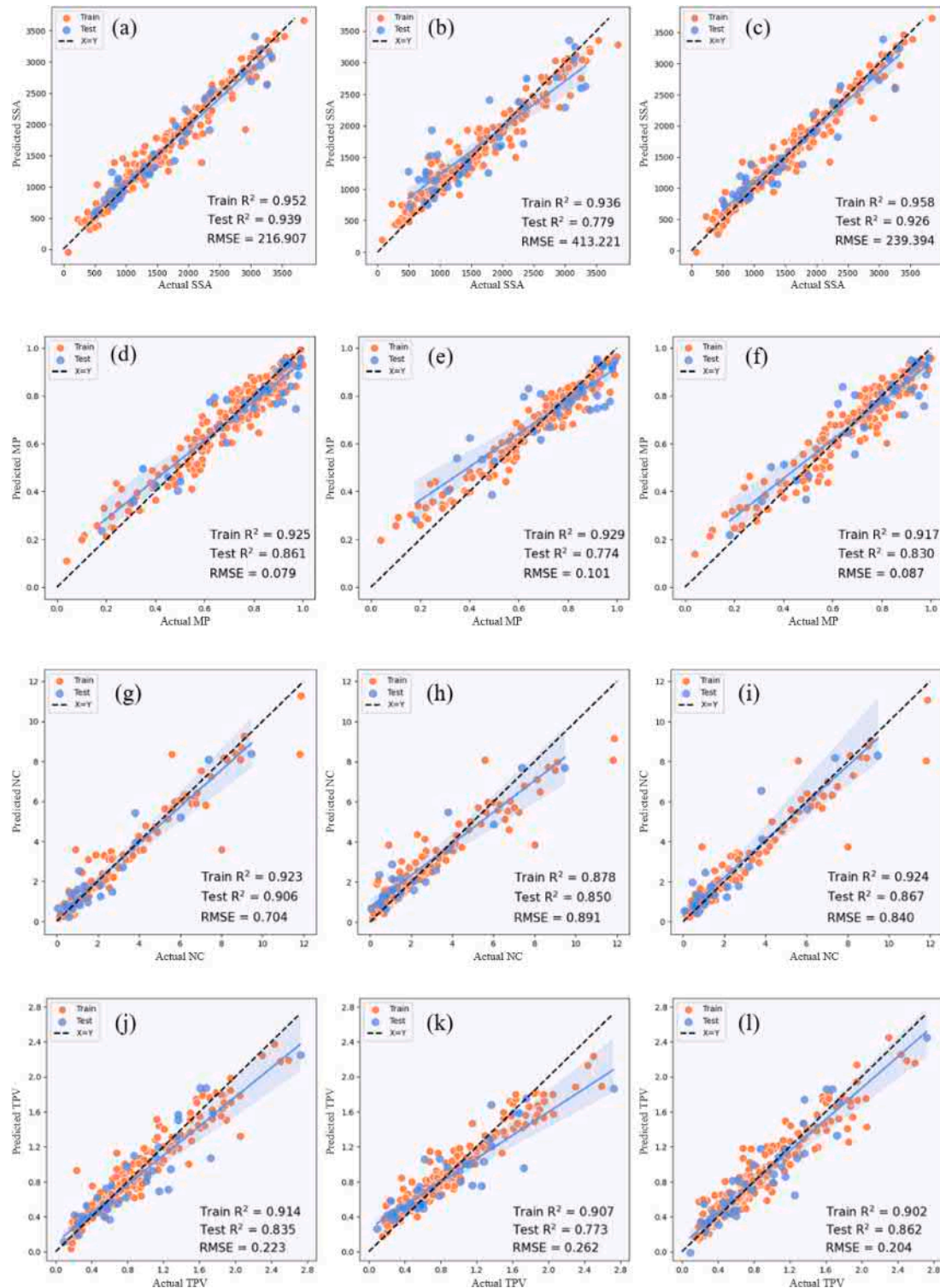


Fig. 4. Comparison of predicted and actual values using GBDT, RF and XGB models (a)(d)(g)(j), GBDT; (b)(e)(h)(k), RF; (c)(f)(i)(l), XGB models.

testing set was not as good as that in the training set, indicating that the prediction accuracy of the testing set was lower than that of the training set. In fact, it is normal that the prediction accuracy of the trainset is always higher than that of the test set using the ML model [13]. The R^2 of the model evaluation metrics also directly demonstrates the similar idea, which reveal that the R^2 of the ML models based on GBDT, RF and XGB are 0.952, 0.936 and 0.958 for the training set, and 0.939, 0.779 and 0.926 for the testing set. Both the GBDT and XGB models with the RMSE scores of 216.9 and 239.3 showed superior prediction accuracy than the RF model with the RMSE of 413.2. In general, the GBDT model displays the greatest predictive accuracy for the specific surface area.

Fig. 4(d), (e) and (f) shows the scatter plot of microporosity. The Figure showed that the R^2 using the three models on the training set were in the range of 0.91–0.93, with the RF model presenting the highest R^2 (0.929), just slightly higher than the GBDT and XGB models. In contrast, the R^2 of the model on the testing set was in the range of 0.77–0.86, all of which were lower than these on the training set, and it was worth noting that the RF model presented the lowest performance on the testing set (0.774), indicating that overfitting occurred using the RF model. The RF model performed excellently in the training set, resulting in poor performance in the validation and testing set. RMSE lie in the range of 0.079–0.101 on the testing set, with errors all within reasonable limits. Another notable finding was that the microporosity was distributed around 0.8, indicating that the hierarchical porous carbon with rich microporous structures could be obtained via biomass pyrolysis and activation.

Fig. 4(j), (k) and (l) shows the scatter plot of the total pore volume of porous carbon. All three models achieved high prediction accuracy ($R^2 > 0.9$) on the training set, with the highest R^2 of 0.914 using the GBDT model, in which the dataset points were more concentrated near the

black dashed line. In the testing set, largest R^2 (0.862) and the smallest RMSE (0.204) are obtained using the XGB model. The distribution of blue dots for the XGB model was closer to the black dashed line compared to the other two models, indicating that the XGB model advantageous in predicting the total pore volume of porous carbon.

Fig. 4(g), (h) and (i) provides the scatter plot of N content in porous carbon. Again, both the GBDT and XGB models presented the better scores than the RF model on both the training and testing sets, with R^2 of 0.923 (GBDT), 0.924 (XGB) and 0.878 (RF) on the training set, and R^2 of 0.906 (GBDT), 0.867 (XGB) and 0.850 (RF) on the testing set, respectively. The RMSE using GBDT model is 0.704, which was significantly lower than the other two models of 0.891 (RF) and 0.840 (XGB). Hence, GBDT presented better prediction accuracy and generalization ability than others.

In general, both the GBDT model and the XGB model present an excellent prediction on training and testing datasets, and the ultra-strong prediction performance of the XGB model on the training set brings about the overfitting and tuning optimization process more time-consuming and arduous, thus the GBDT model is used below to analyze the feature contribution of various prediction targets.

3.4. Feature contribution using GBDT model

The GBDT model showed better performance than the other two models for all four prediction targets, indicating that the GBDT model could better describe the multidimensional and complex relationships between the input features and output features, the GBDT model is applied in this study to explore the contribution of each input feature. Fig. 5 provides the contributions of individual input features and each group to output features using the GBDT model and the detailed values

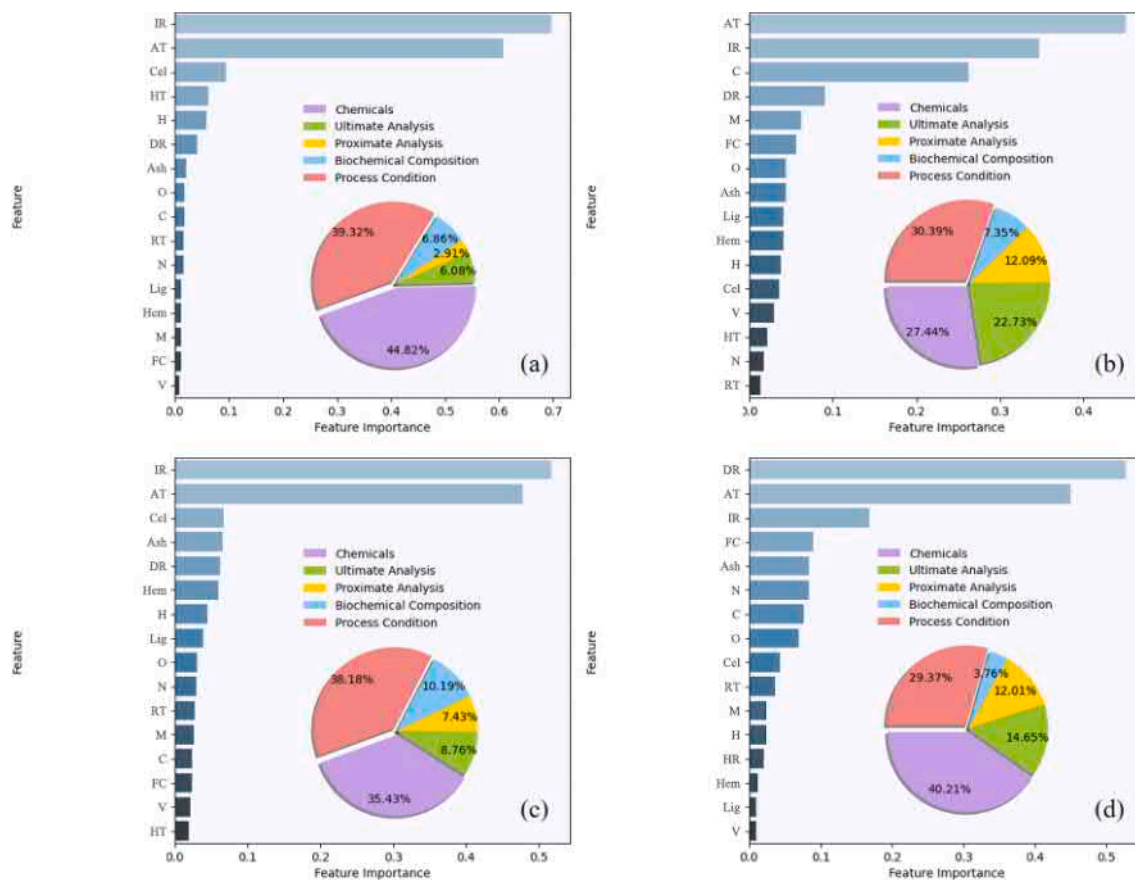


Fig. 5. Contribution of individual input features and each group using the GBDT model (a) specific surface area, (b) microporosity, (c) total pore volume and (d) N content.

of the feature contribution are provided in [Supplementary Material \(Table S2\)](#). In terms of the contribution of individual features, it can be seen that the activation temperature, nitrogen-doped ratio and impregnation ratio are extremely important factors in predicting the four different target features (specific surface area, total pore volume, microporosity and nitrogen content) of porous char, and are in the first echelon. Furthermore, in terms of the group, chemicals and activation conditions together accounted for 60–70% of all contributions, which was sufficient to demonstrate the significance of these two influencing factors in the preparation of biomass-derived hierarchically porous carbon. Under the activator and high temperature, complex reactions such as carbon lattice expansion and metal intercalation can occur inside porous carbon [35], which can effectively promote the formation of porous structures and increase its surface area. Therefore, many researchers proposed the coordinated adjustment of the activation temperature and activating agent to achieve the better pore formation.

In the predicted results of the specific surface area in [Fig. 5\(a\)](#), the single feature of impregnation ratio is ranked first, reflecting the great influence of the amount of activating agent on the specific surface area. Due to the etching effect of KOH as the most popular activating agent, the carbon is converted into carbonate by acting as an oxidizing agent, pores are developed in the carbon skeleton. Determining the optimal impregnation ratio is particularly important since it will result in varying degrees of porosity, and carbon skeleton may collapse due to excessive activating agent [36]. In the preparation of hierarchical porous carbon, determining the optimal impregnation ratio for various activating agent is also a popular research issue [37,38]. The activation temperature was ranked second, which play nearly the same role as the activating agent. With the increasing activation temperature, the pores formed expands further, and the specific surface area, the total pore volume and average pore size increase. With the extremely high activation temperature, it led to the collapse of the pore structure or shrinkage, leading to a decrease in porosity [36]. Therefore, the activation temperature had a significant effect on the morphological structure of carbon. The cellulose content is ranked third. The macromolecular polysaccharide is composed of glucose and contains many intermolecular hydrogen bonds that contribute to the formation of mesopores [39]. Therefore, the amount of cellulose content also has a substantial degree of influence on the pore structure of carbon. Among all the input feature contributed to the specific surface area, the heating rate was ranked fourth. Sun et al. revealed that the heating rate plays a vital role on the specific surface area and the distribution of pores [40]. Among all the groups, the influence of chemicals is the most significant, followed by the activation conditions, then the elemental composition of the biomass feedstock, the industrial composition and the biochemical composition, and all three of which account for only a tiny share.

A similar pattern was shown in the predicted results of total pore volume and microporosity, as shown in [Fig. 5\(b\) and \(c\)](#), in which the contribution of the predicted results of total pore volume was ranked followed impregnation ratio, activation temperature, cellulose content, ash and nitrogen-doped ratio. In terms of microporosity, it was important to note that the nitrogen-doped ratio is also very significant. NH_3 and its free radicals (NH_2^* , NH^* , and H^*) was greatly produced by the decomposition of various amounts of nitrogen-doped sources during the heating around 500 °C, and they reacted with the oxygen-containing functional group at the micropores by weak chemical etching, inducing the decomposition of the carbon [41,42].

In the prediction of nitrogen content in porous carbon as shown in [Fig. 5\(d\)](#), the nitrogen-doped ratio contributed the largest, which is a matter of course. After nitrogen doping, the acid/base sites can be provided in the porous carbon to improve its electrical conductivity and electrochemical property, and also the micro structure of porous carbon can be easily adjusted, and more active sites are provided for gas adsorption. The activation temperature and the impregnation ratio used both played a significant role. In the process of nitrogen doping, the nitrogen-doped source reacts with the active material in the biochar

(primarily oxygen-containing functional groups) to produce nitrogen-containing functional groups. In further high-temperature activation, the active material is destroyed due to the excessive etching effect, including the nitrogen-containing functional groups. Therefore, keeping the activation temperature and the impregnation ratio at an equilibrium state will be crucial. The collected lignocellulosic agricultural waste typically contains less nitrogen (0.6 wt% on average), which is significantly lower than the typical nitrogen-doped ratio, thus it was noteworthy that the self-borne nitrogen as a carbon precursor in the porous carbon has rather less of an impact on the nitrogen content.

In summary, the nitrogen-doped porous carbon (NPC) presents tunable pore size and specific surface area and the N content can be controllable, indicating that the physicochemical property of NPC could be tuned and NPC is potential in numerous applications, and its efficient synthesis has become an important research topic.

3.5. Partial dependence plots of the variables

In addition to the feature contribution, partial dependence plots (PDP) are provided with features that contributed more and then the influence of these features on the prediction targets is analyzed. In terms of the surface area and total pore volume as shown in [Fig. 6\(a\) and \(b\)](#), they showed an initial increasing trend with the rising activation temperature and then they decrease after reaching 800 °C, as reported in previous experimental studies. Ma et al. prepared nitrogen-doped porous carbon from potato waste residue (PWR) as the carbon source, zinc chloride (ZnCl_2) as the activator, and melamine as the nitrogen dopant, the effect of the activation temperature between 600 and 900 °C on the physicochemical property of porous carbon was investigated. The gravelly particles and a partially porous structure without any defined shape was included in the porous carbon prepared at 600 °C. In contrast, the porous carbon prepared at 700 °C exhibited the larger network structure and the distinct porous morphology appeared, reaching the maximum specific surface of 1052 m^2/g and the maximum pore volume of 0.61 cm^3/g , which could be attributed to the pore development in the carbon structure at the higher carbonization temperature. However, when the activation temperature reached 800 °C, the reticular structure decreased gradually and even disappeared at 900 °C, and the specific surface area remaining only 697 m^2/g at 900 °C [43]. The specific surface area and total pore volume showed a trend of increasing and subsequently dropping with the increasing impregnation ratio. There was a rising trend with the impregnation ratio in the range of 1–3 and a declining trend at around 3–5. With too much activating agent, the more violent reaction occurs, and the carbon skeleton collapse and the pore structure of the carbon is destroyed. However, it returns to larger values when the impregnation ratio reaching above 6, which may be due to the inclusion of some of the less active activators, such as ZnCl_2 and KHCO_3 , in the input database. The increase in the cellulose content leads to the decreasing specific surface area and total pore volume. Mark Gale et al. reported that the hydroxyl group in the cellulose reacted with KOH and the micropores formed was destroyed, leading to a decrease in surface area and total pore volume [44]. The heating rate still has a relatively important effect on the specific surface area. From lower to higher heating rates, the activation process becomes more and more intense, and the pore walls became thinner, causing the transformation of carbon from poorly connected macropores to a hierarchical porous structure. When the heating rate exceeds 10 °C/min, the violent reaction between the activator and the carbon material could lead to the aggregation of micropores and the appearance of mesopores. Therefore, the specific surface area gradually decreases. The heating rate of 10 °C/min is used by most researchers to ensure the thorough activation as well as the test time and energy consumption. Considering the effect of ash on the total pore volume, it increases with the increasing ash and reached the maximum at 15%.

[Fig. 6\(c\)](#) shows the partial dependence plot of some input features on the nitrogen content. It showed a monotonic decreasing trend with the

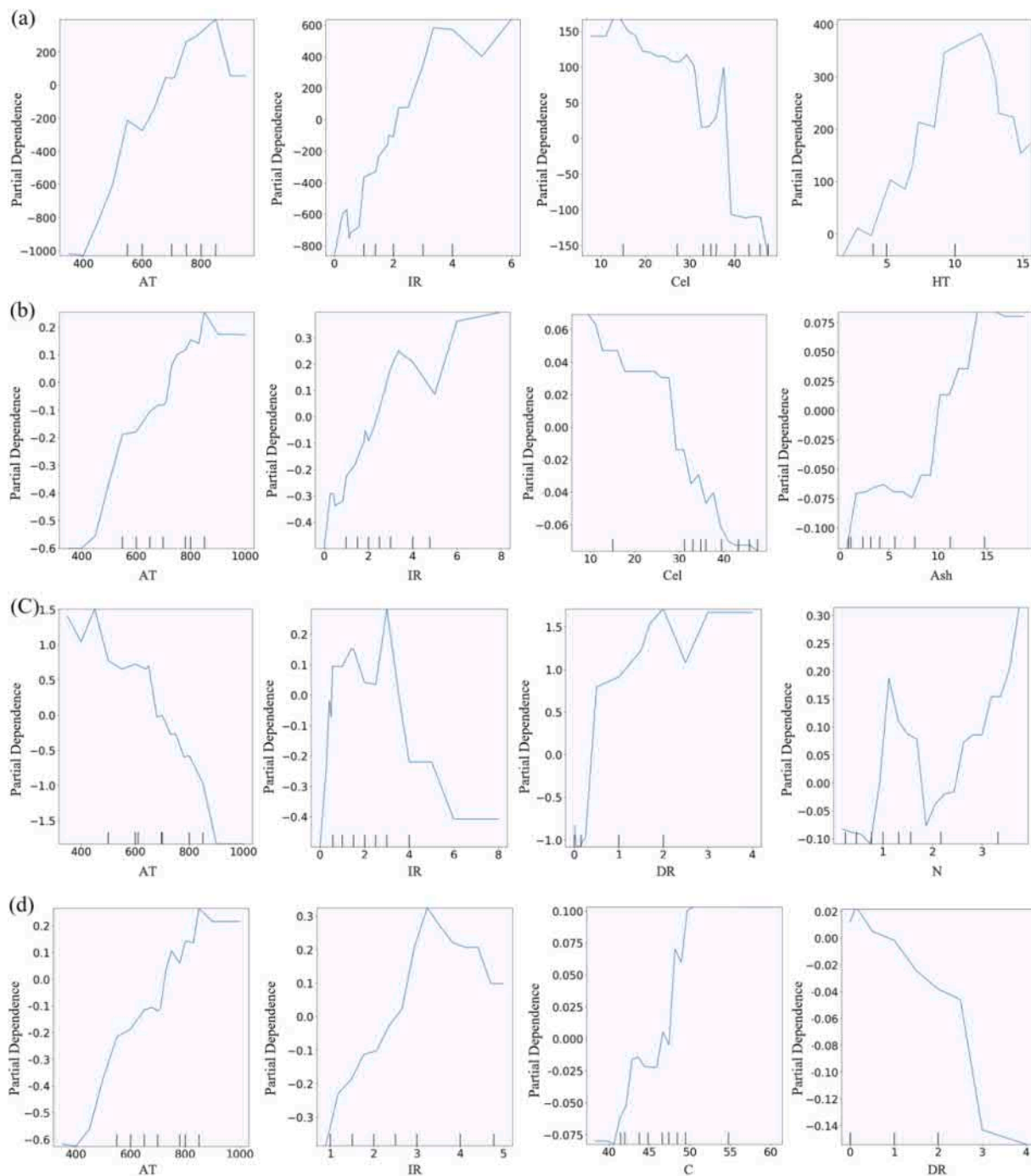


Fig. 6. Partial dependence analysis plots of top 4 input features on each prediction target (a) SSA, (b) TPV, (c) NC and (d)MP.

rising activation temperature. Some N-containing functional groups were destroyed by excessive corrosion, thus reducing the fixation of nitrogen in the porous carbon [42]. Taking the impregnation ratio account, the nitrogen content in the porous carbon increases, then decreased as the impregnation ratio increased, and it peaks at 3. A certain quantity of activating agent was necessary to ensure that the biochar surface is etched away thoroughly and numerous active sites are exposed. The subsequently added nitrogen-doped source in the form of NH_3 reacts greatly with these oxygen-containing functional groups to be immobilized as nitrogen-containing functional groups. However, with the extremely high impregnation ratio, the biochar may be eroded excessively and the nitrogen-containing functional groups may be destroyed, lowering the nitrogen content [45]. In the PDP diagram of the

nitrogen-doped ratio, the nitrogen content in the porous carbon gradually increases with the rising nitrogen doping. The interaction between the active substance and the nitrogen source reached saturation, and the nitrogen content fixed when the nitrogen doping ratio close to 2. It should be pointed out that it is not the higher nitrogen doping ratio, the higher nitrogen content in the porous carbon [46]. As the N in the carbon precursor, it must have some positive correlation to the nitrogen-doped hierarchical porous carbon formed after activation. Since the nitrogen content in the lignocellulosic agricultural wastes in this study was extremely small and negligible, and it was not analyzed here.

In Fig. 6(d), the microporosity increases dramatically as the activation temperature increasing from 400 to 800 °C and the impregnation ratio increasing from 1 to 3, which is in agreement with most of the

literature [47,48]. Moreover, a further increase in the activation temperature and amount of activator led to a more intense reaction, resulting in severe etching effect, thus leading to the larger microporous pore size and the formation of more mesopores. It was notable that the C content of the carbon precursor seemed to have a little synergistic effect on the microporosity. An increased structural aromatization during pyrolysis with the carbon precursor with high C content, forming a regular structure, and part of the carbon structure would shrink again during high-temperature activation, leading to the narrowing of some pores and therefore a continuous increase in microporosity [49]. Considering the effect of nitrogen doping ratio on the microporosity, it showed a monotonic decreasing trend, which may be due to the combination of the doped nitrogen source and the oxygen-containing functional groups, and the microporosity further increases.

4. Conclusion

Using the combined characteristics of biomass feedstock, activation condition and chemicals as inputs and the SSA, TPV, MP and NC of hierarchical porous carbon as outputs via two-step biomass pyrolysis and activation, regression models were developed based on GBDT, RF and XGB. The selection of input features and the setting of the hyper-parameters for these models were both optimized. 7 combination groups of input features were taken into consideration and evaluated, and it revealed that Case1 with all 16 input features, presented the highest score and was utilized for the following prediction. The results showed that GBDT presented the better prediction accuracy and generalization ability than the other two algorithms, with the Test R^2 of 0.939, 0.835, 0.861 and 0.906 for the four prediction targets.

In the analysis of feature contribution of all inputs, all four targets were mainly influenced by the activation temperature and impregnation ratio, and a suitable combination of activating agent and activation temperature will be beneficial to obtain porous carbon with rich pores. The nitrogen-doped ratio presented an extra significant impact on the MP because NH_3 formed by the decomposition of the nitrogen source at high temperature could penetrate deep into the interior space of micropores, realizing the enlargement of pore size. And nitrogen-doped ratio also played a major role in NC during which the nitrogen source could combine with the active material exposed by the activator to fix nitrogen-containing functional groups. The partial dependence analysis revealed that the activation temperature and impregnation ratio acted as the primary influences on the physiochemical property of porous carbon. At the reasonable activation temperature around 800 °C and suitable impregnation ratio around 3, the obtained porous carbon presented an excellent hierarchical morphology and rich pores. At the low nitrogen doping ratio about 2, the appropriate nitrogen doping may be realized.

The synthesis of NPCs for various applications such as (supercapacitors, catalyst carriers, heavy metal adsorbents and gas adsorbents) can be purposefully regulated by constructing the relationship between raw materials, process conditions and the pore structure, N content of NPCs assisted by machine learning.

CRedit authorship contribution statement

Xiaohua Li: Writing – original draft, Methodology, Conceptualization. **Ziheng Huang:** Writing – original draft, Investigation, Formal analysis, Data curation. **Shanshan Shao:** Writing – review & editing, Project administration, Methodology, Conceptualization. **Yixi Cai:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2023.129623>.

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